# POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Digital Logic (New) | Pass Marks: 45
Time :3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) What are advantages and disadvantages of analog and digital systems? Explain. 5
   b) Determine the value of base x if $(50)_x = (203)_4$ . 5
   c) What is weighted and non-weighted code? Why Gray code is called reflected code? 5
2. a) Simplify the given function using K-Map. Also realize the simplified logic circuit using NAND gate only. 8
   $F(A, B, C, D) = \Sigma_m(0, 2, 6, 8, 10) + d = \Sigma_m(11, 12, 13, 14, 15)$
   b) Construct a full subtractor circuit using two half subtractors. Also verify your answer. 7
3. a) Design a combinational circuit that converts 4-bit Excess-3 code into Gray code. 8

**OR**

Design a combinational circuit that converts a decimal digit from the 2421 code to BCD

   b) State and prove De Morgans theorem. Construct F = AC + B'D only using universal gates. 7
4. a) Design and implement BCD to decimal converter circuit using 3-8 linedecoder. 8

**OR**

Implement the following Boolean functions using 3*4*2 PLA.

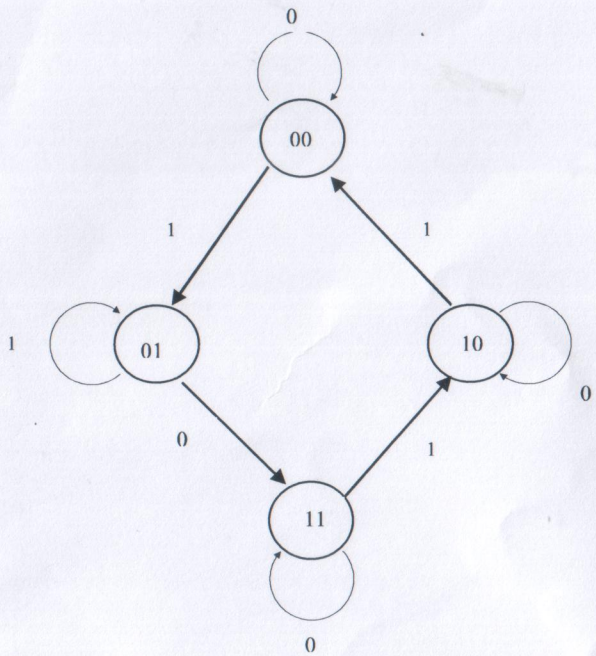
$F1(A, B, C) = \Sigma(3, 5, 6, 7)$

$F2(A, B, C) = \Sigma(0, 2, 4, 7)$

   b) For the Boolean function F(A, B, C) = A+B'C : 7
      i. Obtain the truth table
      ii. Implement it using a MUX
      iii. Implement it using a ROM



5. a) Realize the following state diagram into a sequential circuit using SR fliflop: 8




   b) Define flip flop. Explain the working of JK flip flop with its truth table and logic diagram. Also explain how it eliminates the problem encountered in SR flip flop. 7
6. a) Design 2-bit synchronous up counter by using JK Flipflop. 8
   b) Design a 4-bit arithmetic circuit which performs eight different arithmetic operations. 7
7. Write short notes on: (**Any two**) 2×5
   a) Priority Encoder
   b) Status Register
   c) Master-Slave flip-flop